

Thermal Barrier Imaging Note

A mixed-media methods report for coating YSZ-8

Abstract

Thermal diffusivity was estimated from pulse-heating images of a thermal barrier coating. A spatial boundary condition and a conservative confidence interval were used to separate edge loss from through-thickness transport.

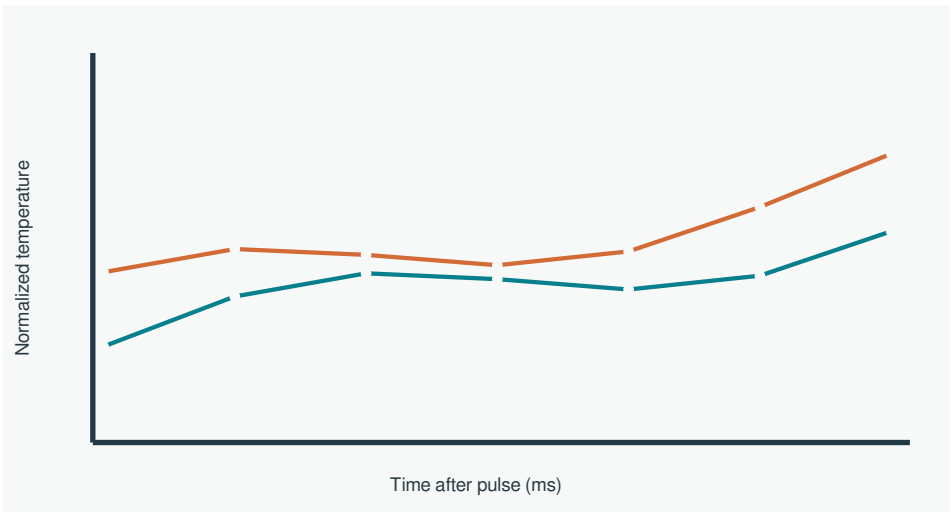
1. Specimen and instrument

Coupons carrying coating YSZ-8 were imaged by IRX-4 at 200 frames s⁻¹. Batch mark TB-17 was recorded before each run, and a dark frame was subtracted without smoothing.

$$\alpha = k / (\rho c_p)$$

(1)

Equation (1) defines thermal diffusivity from conductivity k , density ρ , and heat capacity c_p . Reported uncertainty does not include between-batch coating porosity.



2. Processing sequence

1. Register the IRX-4 frames to the unheated reference without rescaling the specimen.
2. Estimate the boundary condition from the outer 0.5 mm band and retain its sign.
3. Fit thermal diffusivity over 8-42 ms, then compute the confidence interval by profile likelihood.
4. Reject a fit only when residual structure is repeated in three adjacent regions.

$$Fo = \alpha t / L^2 \quad (2)$$

2.1 Fit summary

Region	α (mm ² s ⁻¹)	95 % interval	Status
Center	0.412	0.398-0.427	retained
Edge	0.365	0.331-0.402	retained
Void	0.271	0.188-0.349	flagged

Table 1. Fit results for batch TB-17. Interval limits retain the original decimal precision.

3. Scanned laboratory insert

Printed copy archived with batch TB-17

Operator observation

A faint annulus appeared after the second pulse but did not coincide with the specimen edge. The pattern was retained as an observation because removing it changed the fitted thermal diffusivity by $0.018 \text{ mm}^2 \text{ s}^{-1}$.

Repeat image

The repeat exposure used the same boundary condition. Its confidence interval overlapped the first exposure, although the upper residual band was less uniform.

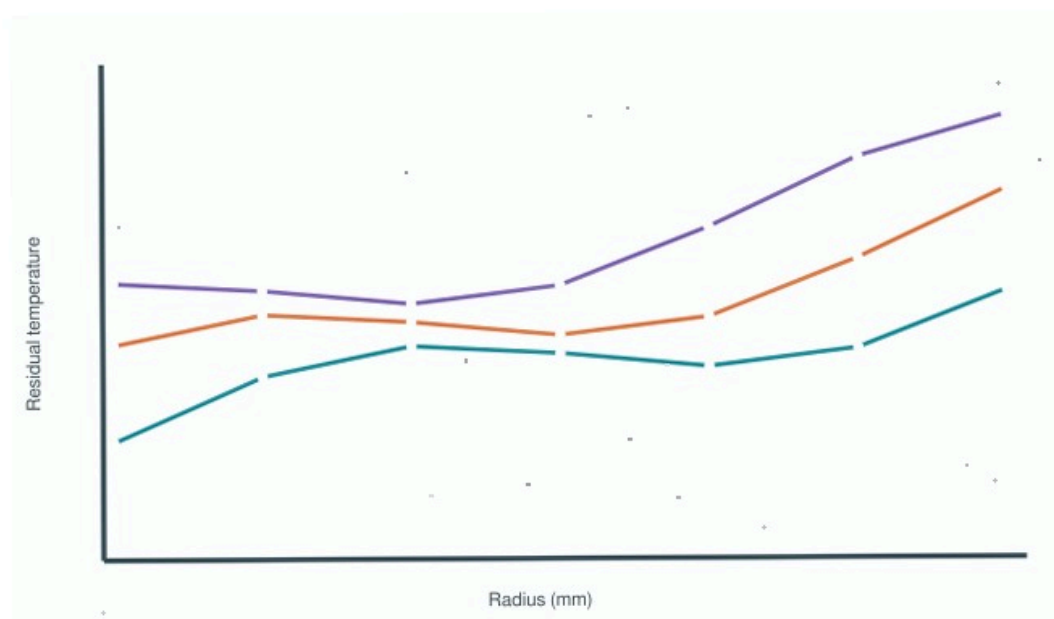


Figure 2. Radial residual profiles from three pulse exposures. TB-17 marks the archived sheet.

TB-17

4. Landscape validation matrix

Cross-region comparison for YSZ-8

Validation summary

The matrix compares fitted and withheld image bands. Confidence intervals are descriptive; they are not a substitute for independent material characterization.

Band	Frames	Boundary condition	α	95 % interval	Decision
A	42	fixed	0.412	0.398-0.427	retain
B	38	estimated	0.405	0.382-0.431	retain
C	41	fixed	0.291	0.206-0.367	review
D	40	estimated	0.399	0.371-0.430	retain

Note. IRX-4 frame timing was verified before the landscape summary was compiled.